

5G RAN

Resource Control in NSA Networking Feature Parameter Description

Issue	Draft A
Date	2025-12-31



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....**1**

2 About This Document.....**3**

2.1 General Statements..... 3

2.2 Features in This Document..... 3

2.3 Differences..... 4

3 Overview.....**5**

4 Resource Control in NSA Networking.....**7**

4.1 Principles..... 7

4.1.1 RFSP-Group-based RRC_CONNECTED UE Number Control..... 7

4.1.1.1 Basic Function..... 7

4.1.1.2 Resource Sharing Between Network Slice Groups and RFSP Groups..... 8

4.1.2 RFSP-Group-based RB Resource Control..... 9

4.1.2.1 Basic Function..... 9

4.1.2.2 Resource Sharing Between Network Slicing Groups and RFSP Groups..... 12

4.1.2.3 Operator-specific Resource Sharing Enhancement..... 13

4.2 Network Analysis..... 16

4.2.1 Benefits..... 16

4.2.2 Impacts..... 16

4.3 Requirements..... 17

4.3.1 Licenses..... 17

4.3.2 Functions..... 17

4.3.2.1 Prerequisite Functions..... 17

4.3.2.2 Mutually Exclusive Functions..... 18

4.3.2.3 Function Impacts..... 20

4.3.3 Cells..... 23

4.3.4 Hardware..... 23

4.3.5 Others..... 24

4.4 Operation and Maintenance..... 24

4.4.1 Data Configuration..... 24

4.4.1.1 Data Preparation..... 24

4.4.1.2 Using MML Commands..... 26

4.4.1.3 Using the MAE-Deployment..... 28

4.4.2 Activation Verification..... 29

4.4.3 Network Monitoring..... 29

5 Parameters..... 30

6 Counters..... 32

7 Glossary..... 33

8 Reference Documents.....34

1 Change History

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between RFSP-group-based RB resource control and flexible experience differentiation. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an mutually exclusive relationship between RFSP-group-based RB resource control and FSD transmission mode. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Revised descriptions of the principles of RFSP-group-based RB resource control. For details, see [4.1.2.1 Basic Function](#) and [4.1.2.3 Operator-specific Resource Sharing Enhancement](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include text-to-table transformation and content position adjustment.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
None	None	4 Resource Control in NSA Networking

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Resource control in NSA networking	Supported only in NSA networking	4 Resource Control in NSA Networking

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Resource control in NSA networking	Supported only in low frequency bands	4 Resource Control in NSA Networking

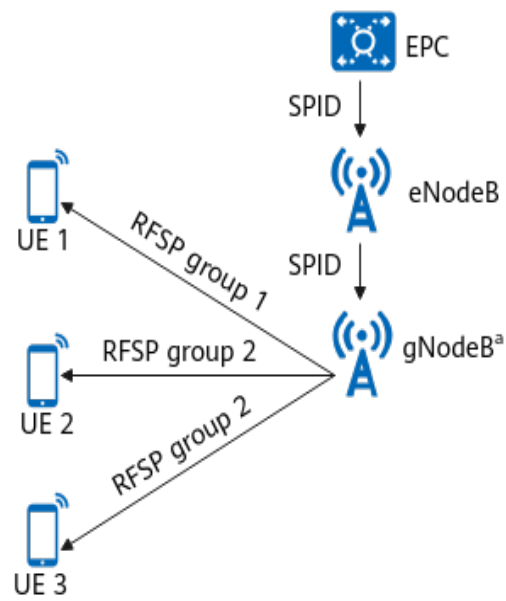
3 Overview

Radio networks provide different radio resource management (RRM) policies to meet the various radio resource requirements of UEs with different service types in different load statuses. As such, RRM policy indexes SPID and RFSP are introduced in LTE and NR networks, respectively. SPID stands for subscriber profile ID (ranging from 1 to 256), which is registered in a home subscriber server (HSS) database. RFSP stands for RAT/Frequency Selection Priority (ranging from 1 to 256), which is registered in a unified data management (UDM) database.

A gNodeB can group UEs with the same characteristics by RFSP (in SA networking) or SPID (in NSA networking) and flexibly customize RRM policies (such as camping, handover, and discontinuous reception) for different UE groups, thereby improving user experience. This function is called flexible user steering. For details, see *Flexible User Steering*.

In NSA networking, the gNodeB can also control the RRC_CONNECTED UE number and RB resources for UEs in different SPID groups to further improve user experience. For this, the evolved packet core (EPC) sends the SPIDs to the eNodeB over the S1 interface, and the eNodeB sends the SPIDs to the gNodeB through the *subscriberProfileIDforRFP* information element (IE) in the SgNB Addition Request message. After mapping the SPIDs to the local configuration, the gNodeB applies specific resource control policies to UEs based on the corresponding SPID groups, thereby better providing targeted services. In NSA networking, the gNodeB processes the SPIDs sent from the eNodeB in the same way as in SA networking. As such, this document uses RFSP to describe RRM policy indexes, as shown in [Figure 3-1](#).

Figure 3-1 RFSP-based grouping



a: SPID is referred to as RFSP on the gNodeB side.

NOTE

In SA networking, network slicing can be enabled to control resources for UEs in different network slice groups. For details, see *Network Slicing*.

In NSA and SA hybrid networking, resource control in NSA networking and network slicing can be enabled together.

4 Resource Control in NSA Networking

4.1 Principles

Resource control in NSA networking involves RRC_CONNECTED UE number and RB resource control based on RFSP groups.

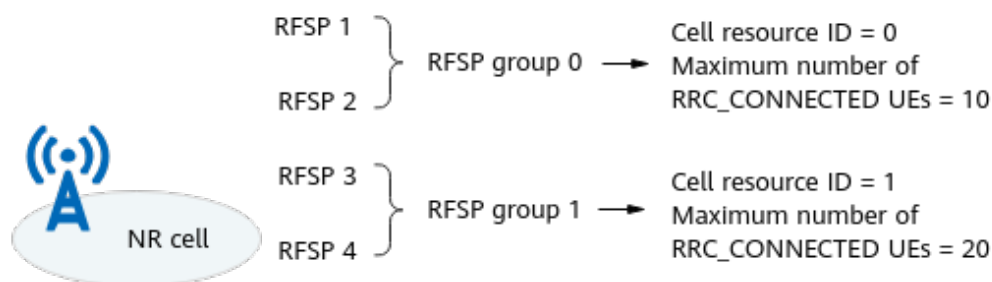
4.1.1 RFSP-Group-based RRC_CONNECTED UE Number Control

The gNodeB supports configuration of the maximum number of RRC_CONNECTED UEs for different RFSP groups, ensuring service experience for UEs in these groups (see [4.1.1.1 Basic Function](#)). In addition, RFSP-group-based RRC_CONNECTED UE number control enables resource sharing between network slice groups and RFSP groups, minimizing configuration specifications for both group types in NSA and SA hybrid networking (see [4.1.1.2 Resource Sharing Between Network Slice Groups and RFSP Groups](#)).

4.1.1.1 Basic Function

The gNodeB supports configuration of the maximum number of RRC_CONNECTED UEs for different RFSP groups, ensuring service experience for UEs in these groups. [Figure 4-1](#) shows an example.

Figure 4-1 Configuration example for RFSP-group-based RRC_CONNECTED UE number control



A maximum of 32 RFSP groups can be configured for each NR cell. You are advised to properly configure the maximum number of RRC_CONNECTED UEs for

each RFSP group of an NR cell based on the RB resources occupied by each RFSP group of the NR DU cell. To configure the maximum number of RRC_CONNECTED UEs for an RFSP group:

1. Configure a cell resource ID for an NR cell and associate the cell resource ID with a maximum number of RRC_CONNECTED UEs. After the configuration, the maximum number of RRC_CONNECTED UEs associated with the cell resource ID is guaranteed for an RFSP group. [Table 4-1](#) describes the related parameters.

Table 4-1 Parameters used to configure the maximum number of RRC_CONNECTED UEs for an NR cell

Parameter	Function
NRCCellRes.NrCellId	Specifies an NR cell.
NRCCellRes.CellResId	Specifies a cell resource ID.
NRCCellRes.MaxRrcConnUserNum	Specifies a maximum number of RRC_CONNECTED UEs.

2. Configure an RFSP group for the NR cell and associate the RFSP group with the cell resource ID. After the configuration, the RFSP group can use the guaranteed maximum number of RRC_CONNECTED UEs. [Table 4-2](#) describes the related parameters.

Table 4-2 Parameters used to configure the maximum number of RRC_CONNECTED UEs for an RFSP group

Parameter	Function
NRCCellOpRfsp.NrCellId	Specifies the NR cell for an RFSP group.
NRCCellOpRfsp.OperatorId^a	Specifies the operator ID for an RFSP group.
NRCCellOpRfsp.RfspGrpCellResId	Specifies a cell resource ID for an RFSP group.
NRCCellOpRfsp.RfspIndex^b	Specifies the index of an RFSP.
a: RFSPs in the same RFSP group must belong to the same operator. b: At least one RFSP can be configured and associated with a cell resource ID for an RFSP group. RFSPs associated with the same cell resource ID for an RFSP group form an RFSP group. Each RFSP group contains a maximum of 36 RFSPs.	

4.1.1.2 Resource Sharing Between Network Slice Groups and RFSP Groups

In NSA and SA hybrid networking, resource sharing between network slice groups and RFSP groups is introduced to minimize configuration specifications and enable

unified management of both. **This function binds a network slice group and an RFSP group for the same operator to an NR cell resource group, allowing the maximum number of RRC_CONNECTED UEs to be shared between them.** However, the gNodeB cannot provide differentiated guarantees for the network slice group and RFSP group that are bound to the same NR cell resource group. If differentiated guarantees are required, you are advised to bind the two groups to separate NR cell resource groups.

NOTE

In NSA and SA hybrid networking, the **gNBOperator.NrNetworkingOption** parameter must be set to **SA_NSA** and the **NRDUCellOp.NrNetworkingOption** parameter for the cell must be set to **SA_NSA** or **INVALID**. For details, see *Cell Management*.

4.1.2 RFSP-Group-based RB Resource Control

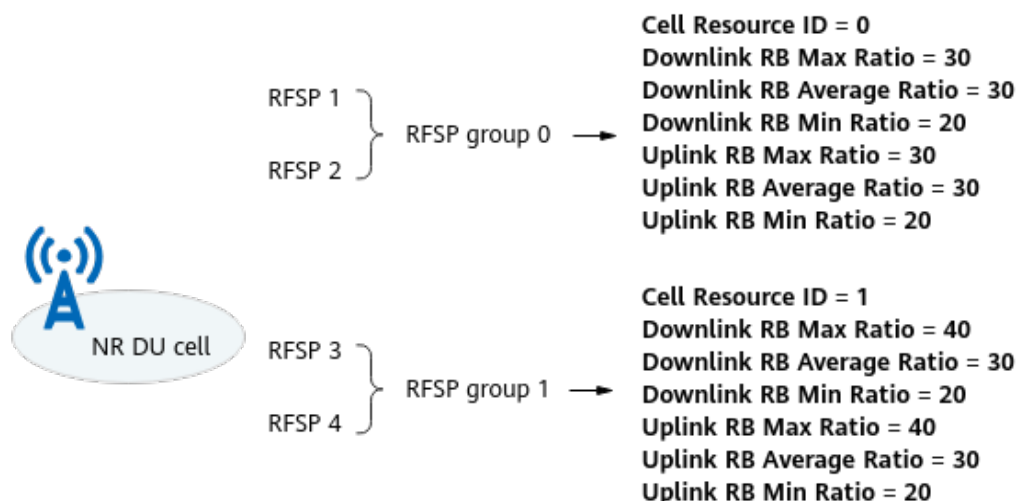
The gNodeB supports configuration and dynamic control of RB resources for different RFSP groups, ensuring service experience for UEs in these groups (see [4.1.2.1 Basic Function](#)). To increase resource allocation accuracy, reduce resource consumption, and further improve user experience, RFSP-group-based RB resource control also enables resource sharing between network slice groups and RFSP groups (see [4.1.2.2 Resource Sharing Between Network Slicing Groups and RFSP Groups](#)) and provides operator-specific resource sharing enhancement (see [4.1.2.3 Operator-specific Resource Sharing Enhancement](#)).

4.1.2.1 Basic Function

The gNodeB supports configuration and dynamic control of RB resources for different RFSP groups, ensuring service experience for UEs in these groups. RB resources refer to PDSCH (downlink) or PUSCH (uplink) resources.

This function is controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw.RfspAlgoSwitch** parameter. The **NRDUCellRes** MO can be used to configure RB resources for different RFSP groups. [Figure 4-2](#) shows an example.

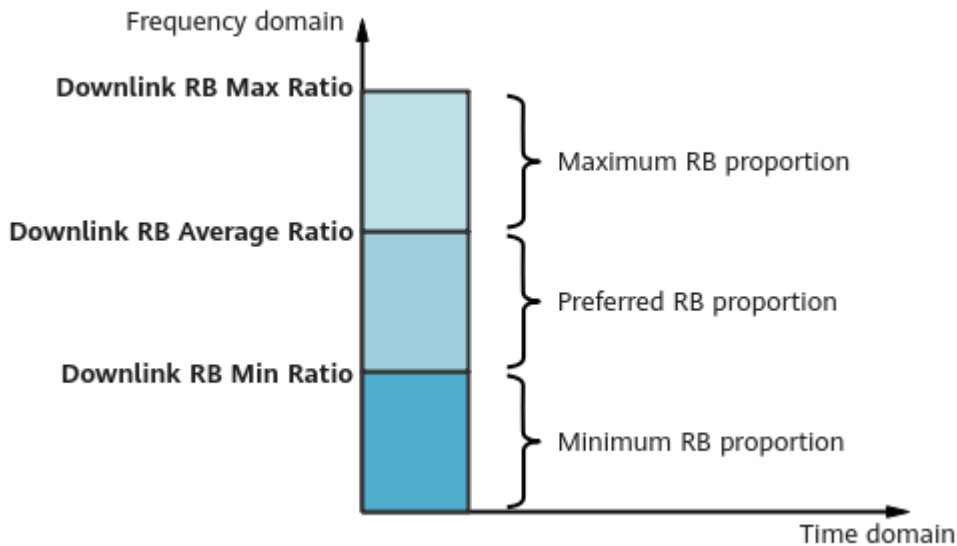
Figure 4-2 Configuration example for RFSP-group-based RB resource control



In the downlink, the PDSCH resources available to UEs in an RFSP group can be controlled by configuring the **Downlink RB Min Ratio**, **Downlink RB Average Ratio**, and **Downlink RB Max Ratio** parameters. These parameters define the minimum, preferred, and maximum RB proportions, as shown in [Figure 4-3](#).

- Minimum RB proportion: The RBs within the quota represented by **Downlink RB Min Ratio** are exclusively used by UEs in the RFSP group.
- Preferred RB proportion: The RBs within the quota represented by **Downlink RB Average Ratio** minus **Downlink RB Min Ratio** are preferentially used by UEs in the RFSP group, and can be used by UEs outside the group when available.
- Maximum RB proportion: The RBs within the quota represented by **Downlink RB Max Ratio** minus **Downlink RB Average Ratio** are not preferentially used by UEs in the RFSP group and are shared with UEs outside the group.

Figure 4-3 Example of downlink RB proportions



A maximum of 32 RFSP groups can be configured for each NR DU cell. To configure RB resources for an RFSP group:

Configure a cell resource ID for an NR DU cell and associate the cell resource ID with RB resources. After the configuration, the RB resources associated with the cell resource ID are guaranteed for an RFSP group. [Table 4-3](#) describes the parameters used to configure RB resources for an RFSP group.

Table 4-3 Parameters used to configure RB resources for an RFSP group

Parameter Name	Parameter ID	Description
NR DU Cell ID	NRDUCellRes.NrDuCellId	Specifies an NR DU cell.

Parameter Name	Parameter ID	Description
Cell Resource ID	NRDUCellRes.CelResId	Specifies a cell resource ID.
Resource Usage Object	NRDUCellRes.ResUsageObject	Specifies the resource usage object. The value RFSP_GROUP of this parameter indicates that the resources are used by an RFSP group.
Uplink RB Min Ratio ^a	NRDUCellRes.UIRbMinRatio	Specifies the proportion of RB resources that can be used only by UEs in an RFSP group to the available PUSCH or PDSCH RB resources in the cell. The minimum proportion of RB resources cannot be used by UEs that are not in the RFSP group.
Downlink RB Min Ratio ^a	NRDUCellRes.DIRbMinRatio	
Uplink RB Average Ratio ^a	NRDUCellRes.UIRbAverageRatio	Specifies the proportion of RB resources that can be preferentially used by UEs in an RFSP group to the available PUSCH or PDSCH RB resources in the cell. If the proportion of RB resources that can be preferentially used are not used up by UEs in the RFSP group, the remaining RB resources can be used by other UEs. In the uplink, the NRDUCellUISch.UIRbResSchPriority parameter can be used to specify the scheduling priority policy of PUSCH RBs for UEs in the RFSP group. • If this parameter is set to NOT_CONFIG, the priority for UEs in the RFSP group is higher than that for UEs with special services and lower than that for signaling messages. • If this parameter is set to HIGH_PRIORITY, the priority for UEs in the RFSP group is higher than that for signaling messages. • If this parameter is set to LOW_PRIORITY, the priority for UEs in the RFSP group is higher than that for UEs with common data services and lower than that for UEs with special services. Special services include voice over NR (VoNR), ultra-reliable low-latency communication (URLLC), and other services that are not categorized as common data services.
Downlink RB Average Ratio ^a	NRDUCellRes.DIRbAverageRatio	

Parameter Name	Parameter ID	Description
Uplink RB Max Ratio ^a	NRDUCellRes. <i>UIRbMaxRatio</i>	Specifies the maximum proportion of RB resources that can be used by UEs in an RFSP group to the available PUSCH or PDSCH RB resources in the cell. Resources scheduled for retransmissions upon discontinuous transmissions (DTXs) are not restricted by the Downlink RB Max Ratio .
Downlink RB Max Ratio ^a	NRDUCellRes. <i>DIRbMaxRatio</i>	

a:

- You can configure one of, or both, the uplink and downlink RB resources, and one, two, or all of the **RB Min Ratio**, **RB Average Ratio**, and **RB Max Ratio**. The downlink parameters must comply with the following configuration rules, which also apply to the uplink.
 - **Downlink RB Max Ratio** ≥ **Downlink RB Average Ratio** ≥ **Downlink RB Min Ratio**
 - In an NR DU cell, the sum of **Downlink RB Min Ratio** configured for all RFSP groups must be less than or equal to 100%.
- These six parameters cannot all be set to **255**.

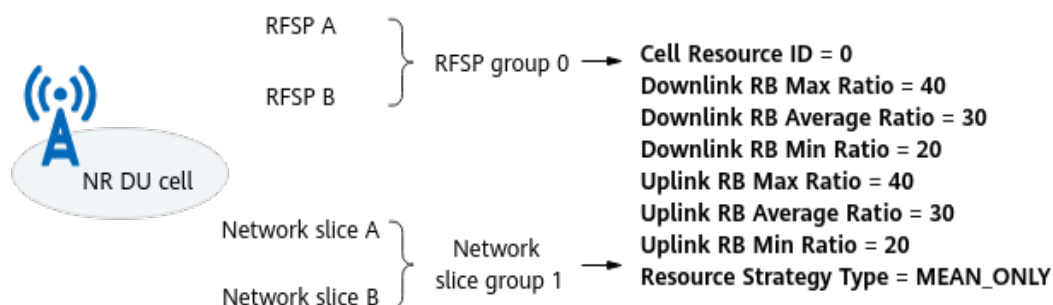
The difference between the proportion of RB resources used by UEs in each RFSP group of an NR DU cell and the configured RB proportion may be greater than 1% when all of the following conditions are met:

- The sum of **Uplink RB Average Ratio** or **Downlink RB Average Ratio** configured for all RFSP groups in the cell is equal to 100%.
- The values of **Uplink RB Average Ratio** or **Downlink RB Average Ratio** configured for all RFSP groups in the cell differ greatly (by more than 70%).
- Each RFSP group is fully loaded with full-buffer services.

4.1.2.2 Resource Sharing Between Network Slicing Groups and RFSP Groups

In NSA and SA hybrid networking, resource sharing between network slice groups and RFSP groups is introduced to minimize configuration specifications and enable unified management of both. When NR DU cell resource management between network slices and RFSP-group-based RB resource control are both enabled and the **NRDUCellRes.ResUsageObject** parameter for the NR DU cell is set to **NS_RFSP_GROUP**, resource sharing between network slice groups and RFSP groups takes effect. As shown in [Figure 4-4](#), this function binds a network slice group and an RFSP group for the same operator to an NR DU cell resource group, whose configurations are shared for scheduling UEs in both groups. In this case, the **NRDUCellRes.ResStrategyType** parameter (specifying the RB resource control policy) can only be set to **MEAN_ONLY**.

Figure 4-4 Configuration example for resource sharing between network slicing groups and RFSP groups



NOTE

For details about network slicing and NR DU cell resource management between network slices, see related sections in *Network Slicing*.

This function does not take effect if either NR DU cell resource management between network slices or RFSP-group-based RB resource control is enabled. In this case, only the enabled function takes effect.

The gNodeB cannot provide differentiated guarantees for the network slice group and RFSP group that are bound to the same NR DU cell resource group. To ensure differentiated service experiences for different UEs, you are advised to bind the two groups to separate NR DU cell resource groups.

4.1.2.3 Operator-specific Resource Sharing Enhancement

In RAN sharing with common carrier, operator-specific resource sharing enhancement is introduced to further increase the accuracy of resource allocation. This ensures fair and flexible operator-specific resource sharing when RFSP-group-based RB resource control is enabled. The function involves intra-operator preferential sharing of idle resources and inter-operator resource sharing policies. Inter-operator resource sharing policies take effect only when intra-operator preferential sharing of idle resources is in effect. The function works the same way in both the uplink and downlink. The following example focuses on the downlink. For parameters and meanings of **Downlink RB Average Ratio** and **Downlink RB Min Ratio**, see [Table 4-3](#). For details about RAN sharing with common carrier, see *Multi-Operator Sharing*.

This section specifically describes operator-specific resource sharing enhancement in NSA networking. For information on operator-specific resource sharing enhancement in SA networking, see *Network Slicing*. In NSA and SA hybrid networking, the RFSP group mentioned here is equivalent to either a network slice group or an RFSP group. [Table 4-4](#) describes the concepts related to this function.

Table 4-4 Related concepts

Concept	Definition	Description
Idle resources	Resources that are not used by UEs in an RFSP group within the quota ranging from Downlink RB Min Ratio to Downlink RB Average Ratio specified for the RFSP group	If Downlink RB Average Ratio is set to 255 , there are no idle resources.
Common resources	All remaining resources beyond the quota represented by the sum of the larger of either Downlink RB Average Ratio or Downlink RB Min Ratio specified for all RFSP groups (or all network slice groups and RFSP groups in NSA and SA hybrid networking) in a cell	If Downlink RB Average Ratio or Downlink RB Min Ratio is set to 255 , the corresponding RB proportion is 0.
Operator-specific shares	Sum of the larger of either Downlink RB Average Ratio or Downlink RB Min Ratio specified for all RFSP groups (or all network slice groups and RFSP groups in NSA and SA hybrid networking) of an operator	If Downlink RB Average Ratio or Downlink RB Min Ratio is set to 255 , the corresponding RB proportion is 0.

Intra-Operator Preferential Sharing of Idle Resources

This function allows idle resources to be preferentially shared among UEs served by the same operator. Any remaining idle resources can then be shared with UEs served by other operators according to the inter-operator resource sharing policies described in [Inter-Operator Resource Sharing Policies](#).

This function takes effect when the **NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePol** parameter is set to **SAME_OPERATOR_PREFERRED**. If this parameter is set to **FAIR_SHARING**, this function does not take effect, and idle resources are allocated based on UE scheduling priorities.

Inter-Operator Resource Sharing Policies

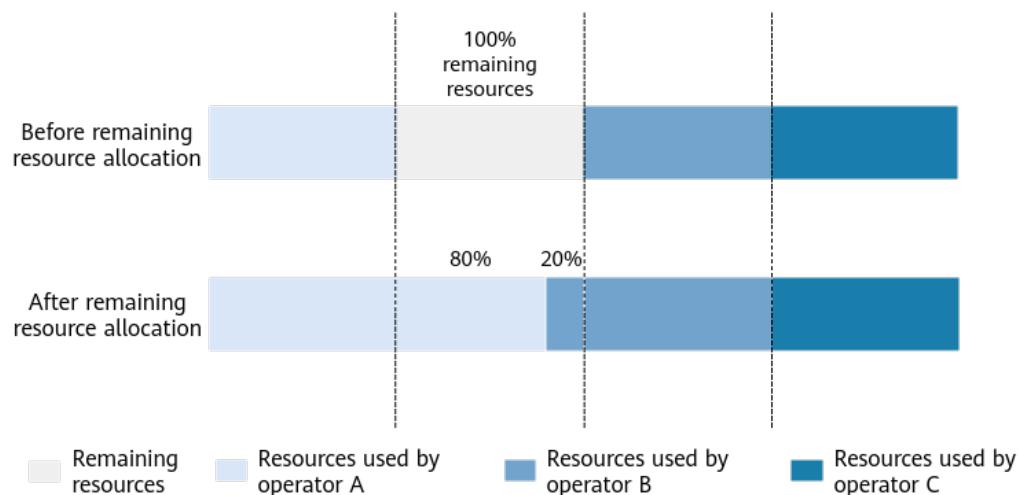
Inter-operator resource sharing policies apply only when the **NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePol** parameter is set to **SAME_OPERATOR_PREFERRED**. In this case, the total remaining resources (including the remaining idle resources after intra-operator preferential sharing and common resources) are shared among UEs served by all operators using specific sharing policies. Whether resources are shared based on the average RB proportion or the number of active UEs is determined by the **NRDUCellNsMgmtConfig.NetSliceRfspGrpPreemptPol** parameter.

- If this parameter is set to **BASED_ON_OP_SHARE**, resource sharing based on operator-specific shares takes effect. The gNodeB allocates the remaining resources in proportion to each operator's share.

As shown in [Figure 4-5](#), if the total average RB proportions configured for all RFSP groups of operators A, B, and C in a cell are 40%, 10%, and 0%, respectively, operator A has priority access to 80% (4/5) of the remaining resources, operator B to 20% (1/5), and operator C has no priority access to the remaining resources.

Figure 4-5 Example of resource sharing based on operator-specific shares

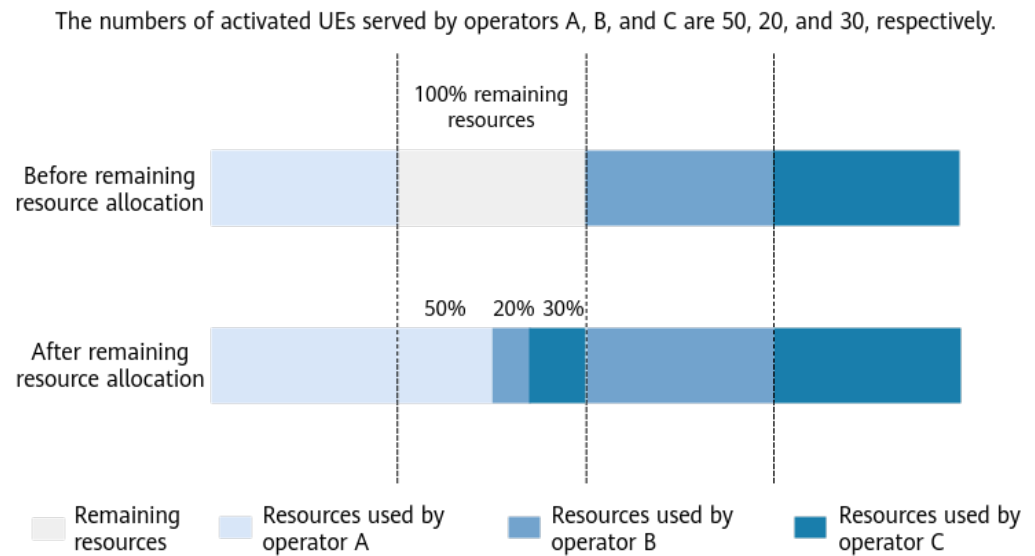
The total average RB proportions configured for operators A, B, and C are 40%, 10%, and 0%, respectively.



- If this parameter is set to **BASED_ON_OP_ACTIVE_USER**, resource sharing based on active UE number takes effect. The gNodeB allocates the remaining resources in proportion to each operator's active UE number.

As shown in [Figure 4-6](#), if the numbers of active UEs served by operators A, B, and C in a cell are 50, 20, and 30, respectively, operator A has priority access to 50% (1/2) of the remaining resources, operator B to 20% (1/5), and operator C to 30% (3/10).

Figure 4-6 Example of resource sharing based on active UE number



NOTE

Active UEs are RRC_CONNECTED UEs with non-empty buffer status reports (BSRs).

When an inter-operator resource sharing policy takes effect and the **NRDUCellRes.ResUsageObject** parameter is set to **NETWORK_SLICE_GROUP**, **NS_RFSP_GROUP**, or **RFSP_GROUP**, the sum of the following results must not exceed 100%. This applies to both the uplink and downlink. Here uses the downlink as an example.

- Sum of all records with **Downlink RB Average Ratio** set to a value other than 255
- Sum of all records with **Downlink RB Average Ratio** set to 255 and **Downlink RB Min Ratio** set to a value other than 255

4.2 Network Analysis

4.2.1 Benefits

This function enables resource control based on RFSP groups in NSA networking to meet the requirements for differentiated network services in NSA networking.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Operator-specific resource sharing enhancement has the following network impacts:

- After intra-operator preferential sharing of idle resources is enabled, the uplink and downlink cell throughputs (**Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**) fluctuate. For an

operator with more RBs allocated, the uplink and downlink UE throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) increase, improving user experience. For an operator with fewer RBs allocated, the uplink and downlink UE throughputs decrease, negatively affecting user experience.

- When inter-operator resource sharing policies take effect, the uplink and downlink cell throughputs fluctuate. For an operator with a large proportion of resources allocated, the uplink and downlink UE throughputs increase, improving user experience. For an operator with a small proportion of resources allocated, the uplink and downlink UE throughputs decrease, negatively affecting user experience.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Intra-operator preferential sharing of idle resources	NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePol set to SAME_OPERATOR_PREFERRED	<i>Resource Control in NSA Networking</i>	Inter-operator resource sharing policies (specified by the NRDUCellNsMgmtConfig.NetSliceRfspGrpPreemptPol parameter) depend on intra-operator preferential sharing of idle resources.
Low - frequency TDD	RFSP-group-based RB resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw.RfspAlgoSwitch parameter	<i>Resource Control in NSA Networking</i>	Intra-operator preferential sharing of idle resources (with the NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePol parameter set to SAME_OPERATOR_PREFERRED) depends on RFSP-group-based RB resource control.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific RB management	RB_DYNAMIC_SHARING_SW and MAX_RB_LIMIT_SW options of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Operator-specific RB management cannot work with RFSP-group-based RB resource control.
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Operator-specific guarantee on the minimum number of available RBs cannot work with RFSP-group-based RB resource control.
Low-frequency TDD	Dynamic RB control for UEs requiring low latency and high reliability	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter	<i>URLLC</i>	Dynamic RB control for UEs requiring low latency and high reliability cannot work with RFSP-group-based RB resource control.
Low-frequency TDD	Cell Combination	HYPER_CELL_COMBINE_MODE option of the NRDUCell.NrDuCellNetworkingMode parameter with NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	Cell Combination cannot work with RFSP-group-based RB resource control.
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	User PRB control for FWA cannot work with RFSP-group-based RB resource control.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI group	NRDUCellRes.ResUsageObject set to QCI_GROUP	<i>QCI-based Resource Control</i>	Intra-operator preferential sharing of idle resources (with the NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePool parameter set to SAME_OPERATOR_PREFERRED) does not work in QCI groups.
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	QCI-based resource control cannot work with intra-operator preferential sharing of idle resources (with the NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePool parameter set to SAME_OPERATOR_PREFERRED).
Low-frequency TDD	FSD transmission mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL and NRDUCellMultiTrp.DataTransMode set to FSD_MODE	<i>Hyper Cell</i>	RFSP-group-based RB resource control cannot work in FSD transmission mode.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink resource isolation	DL_EMBB_PREFERENTIAL_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	If the Downlink RB Min Ratio parameter in RFSP-group-based RB resource control takes effect for FWA users, downlink resource isolation no longer takes effect for the users. If the Downlink RB Min Ratio parameter in RFSP-group-based RB resource control takes effect for non-FWA users, the average downlink throughput of FWA users decreases more severely.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>	When RFSP-group-based RB resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw.RfspAlgorithmSwitch parameter) is enabled together with UL and DL Decoupling and is enabled in the SUL cell, it must also be enabled in the associated TDD cell. Furthermore, the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When RFSP-group-based RB resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw.RfspAlgorithmSwitch parameter) is enabled together with super uplink with independent SUL deployment and is enabled in the SUL cell, it must also be enabled in the associated TDD cell. Furthermore, the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	After both "RFSP-group-based RB resource control" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of the RFSP resource group to which the UE belongs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

4.3.3 Cells

RFSP-group-based RB resource control:

If the **NRDUCell.DuplexMode** parameter of a cell is set to **CELL_SUL** and the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw.RfspAlgoSwitch** parameter corresponding to the **NRDUCellSulRelation.NrDuCellId** parameter of the cell is deselected, the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw.RfspAlgoSwitch** parameter of this cell must be deselected.

4.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Only the UMPTg series and later boards support a maximum of 32 RFSP groups, while other main control boards are limited to six RFSP groups.

RF Modules

This function does not depend on RF modules.

4.3.5 Others

SPIDs must have been configured for UEs on the evolved packet core (EPC) side.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-5 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCeIlRes.NrCeIlId	Set this parameter based on the network plan.
Cell Resource ID	NRCeIlRes.CeIlResId	Set this parameter based on the network plan.
Max RRC Connected User Number	NRCeIlRes.MaXRrcConnUserNum	Set this parameter based on the network plan.
NR Cell ID	NRCeIlOpRfsp.NrCeIlId	The value of this parameter must be identical to the value of NRCeIlRes.NrCeIlId .

Parameter Name	Parameter ID	Setting Notes
Index to RAT/ Frequency Selection Priority	NRCellOpRfsp.RfspIndex	Set this parameter based on the network plan.
Operator ID	NRCellOpRfsp.OperatorId	Set this parameter based on the network plan.
RFSP Group Cell Resource ID	NRCellOpRfsp.RfspGrpCellResId	The value of this parameter must be identical to the value of NRCellRes.CellResId .
RFSP-based Algorithm Switch	NRDUCellCommonSw.RfspAlgoSwitch	Select the RB_DYNAMIC_CONTROL_SW option of this parameter.
NR DU Cell ID	NRDUCellRes.NrDuCellId	Set this parameter based on the network plan.
Cell Resource ID	NRDUCellRes.CellResId	Set this parameter based on the network plan.
Resource Usage Object	NRDUCellRes.ResUsageObject	Set this parameter to RFSP_GROUP or NS_RFSP_GROUP based on the network plan. If this parameter is set to NS_RFSP_GROUP , the NRDUCellRes.ResStrategyType parameter can only be set to MEAN_ONLY .
Downlink RB Max Ratio	NRDUCellRes.DlRbMaxRatio	Set this parameter based on the network plan.
Downlink RB Average Ratio	NRDUCellRes.DlRbAverageRatio	Set this parameter based on the network plan.
Downlink RB Min Ratio	NRDUCellRes.DlRbMinRatio	Set this parameter based on the network plan.
Uplink RB Max Ratio	NRDUCellRes.UlRbMaxRatio	Set this parameter based on the network plan.
Uplink RB Average Ratio	NRDUCellRes.UlRbAverageRatio	Set this parameter based on the network plan.
Uplink RB Resource Scheduling Priority	NRDUCellUISch.UlRbResSchPriority	Set this parameter based on the network plan.
Uplink RB Min Ratio	NRDUCellRes.UlRbMinRatio	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellOpRfsp.NrDuCellId	Set this parameter based on the network plan.
Index to RAT/ Frequency Selection Priority	NRDUCellOpRfsp.RfspIndex	Set this parameter based on the network plan.
Operator ID	NRDUCellOpRfsp.OperatorId	Set this parameter based on the network plan.
RFSP Group Cell Resource ID	NRDUCellOpRfsp.RfspGrpCellResId	Set this parameter based on the network plan.
Network Slice/RFSP Group Sharing Policy	NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePol	Set this parameter based on the network plan. To enable both RFSP-group-based RB resource control and operator-specific resource sharing enhancement, you are advised to set this parameter to SAME_OPERATOR_PREFERRED .
Network Slice/RFSP Group Preemption Policy	NRDUCellNsMgmtConfig.NetSliceRfspGrpPreemptPol	Set this parameter based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

To enable RFSP-group-based RRC_CONNECTED UE number control:

```
//Enabling RFSP-group-based RRC_CONNECTED UE number control
//Configuring the maximum number of RRC_CONNECTED UEs for an NR cell
ADD NRCELLRES: NrCellId=7, CellResId=0, MaxRrcConnUserNum=20;
//Configuring an RFSP group for the NR cell and associating the group with the maximum number of RRC_CONNECTED UEs
ADD NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=1, RfspGrpCellResId=0;
ADD NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=2, RfspGrpCellResId=0;
```

To enable resource sharing between network slice groups and RFSP groups for an NR cell: (The parameters related to network slice deployment on the gNodeB must be configured before configuring a slice in a network slice group for an NR cell. For details about network slice deployment, see *Network Slicing*.)

```
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=1, MaxRrcConnUserNum=10;
//Configuring a network slice group and an RFSP group for the NR cell and binding the NR cell resources to the groups
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=0, CellResId=1, NsGrpDescription="GRP_CU";
ADD NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=3, RfspGrpCellResId=1;
//Configuring a slice in the network slice group for the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
```

To enable RFSP-group-based RB resource control:

```
//Enabling dynamic RB control for the NR DU cell
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring RB resources for the NR DU cell
ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=RFSP_GROUP,
ResStrategyType=MEAN_ONLY, DIRbMaxRatio=30, DIRbAverageRatio=30, DIRbMinRatio=20,
UIRbMaxRatio=30, UIRbAverageRatio=30, UIRbMinRatio=20;
//Configuring an RFSP group for the NR DU cell and associating the group with the RB resources
ADD NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=1, RfspGrpCellResId=0;
ADD NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=2, RfspGrpCellResId=0;
//Configuring the uplink scheduling priority policy for UEs in an RFSP group in the NR DU cell
MOD NRDUCELLULSCH: NrDuCellId=7, UIRbResSchPriority=NOT_CONFIG;
```

To enable resource sharing between network slice groups and RFSP groups for an NR DU cell: (The parameters related to network slice deployment on the gNodeB must be configured before configuring a slice in a network slice group for an NR cell. For details about network slice deployment, see *Network Slicing*.)

```
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources and enabling intra-operator preferential sharing of idle resources and resource sharing based on active UE number
ADD NRDUCELLRES: NrDuCellId=7, CellResId=1, ResUsageObject=NS_RFSP_GROUP, DIRbMaxRatio=10,
DIRbMinRatio=10, DIRbAverageRatio=10, UIRbMaxRatio=10, UIRbMinRatio=10, UIRbAverageRatio=10;
//Configuring a network slice group and an RFSP group for the NR DU cell and binding the NR DU cell resources to the groups
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0, CellResId=1, NsGrpDescription="GRP_DU";
ADD NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=3, RfspGrpCellResId=1;
//Configuring a slice in the network slice group for the NR DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1,
SliceDifferentiator=6543;
```

To enable operator-specific resource sharing enhancement after RFSP-group-based RB resource control or resource sharing between network slice groups and RFSP groups for an NR DU cell is enabled:

```
MOD NRDUCELLNSMGMTCONFIG: NrDuCellId=7, NetSliceRfspGrpSharePol=SAME_OPERATOR_PREFERRED,
NetSliceRfspGrpPreemptPol=BASED_ON_OP_ACTIVE_USER;
```

Deactivation Command Examples

To remove the RFSP-group-specific number of RRC_CONNECTED UEs:

```
//Removing the association between the RFSP group of the NR cell and the maximum number of
RRC_CONNECTED UEs
RMV NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=1;
```

```
RMV NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=2;  
//Removing the maximum number of RRC_CONNECTED UEs configured for the NR cell  
RMV NRCELLRES: NrCellId=7, CellResId=0;
```

To disable resource sharing between network slice groups and RFSP groups for an NR cell: (The network slice must be removed from the gNodeB after running the following MML commands. For details about removing the network slice from the gNodeB, see *Network Slicing*.)

```
//Disabling network slicing enhancement  
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-0;  
//Removing the slice from the network slice group for the NR cell  
RMV NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;  
//Removing the network slice group from the NR cell  
RMV NRCELLNSGRP: NrCellId=7, NsGrpId=0;  
//Removing the association between the RFSP group of the NR cell and the maximum number of  
RRC_CONNECTED UEs  
RMV NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=3;  
//Removing the maximum number of RRC_CONNECTED UEs configured for the NR cell  
RMV NRCELLRES: NrCellId=7, CellResId=1;
```

To disable operator-specific resource sharing enhancement:

```
MOD NRDUCELLNSMGMTCONFIG: NrDuCellId=7, NetSliceRfspGrpSharePol=FAIR_SHARING,  
NetSliceRfspGrpPreemptPol=BASED_ON_OP_SHARE;
```

To remove the RFSP-group-specific RB resources:

```
//Disabling dynamic RB control for the NR DU cell  
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-0;  
//Removing the association between the RFSP group of the NR DU cell and the RB resources  
RMV NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=1;  
RMV NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=2;  
//Removing the RB resources configured for the NR DU cell  
RMV NRDUCELLRES: NrDuCellId=7, CellResId=0;
```

To disable resource sharing between network slice groups and RFSP groups for an NR DU cell: (The network slice must be removed from the gNodeB after running the following MML commands. For details about removing the network slice from the gNodeB, see *Network Slicing*.)

```
//Disabling network slicing enhancement  
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-0;  
//Disabling dynamic RB control  
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-0;  
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-0;  
//Removing the slice from the network slice group for the NR DU cell  
RMV NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1,  
SliceDifferentiator=6543;  
//Removing the network slice group from the NR DU cell  
RMV NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0;  
//Removing the association between the RFSP group of the NR DU cell and the RB resources  
RMV NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=3;  
//Removing the RB resources configured for the NR DU cell  
RMV NRDUCELLRES: NrDuCellId=7, CellResId=1;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

You can observe related performance counters to check whether resource control in NSA networking has been activated. If any of the following counters has a non-zero value, resource control in NSA networking has been activated.

- **N.PRBUUsed.UL.RfspGrp0.Avg** to **N.PRBUUsed.UL.RfspGrp31.Avg**
- **N.PRBUUsed.DL.RfspGrp0.Avg** to **N.PRBUUsed.DL.RfspGrp31.Avg**
- **N.User.RfspGrp0.RRCCConn.Avg** to **N.User.RfspGrp31.RRCCConn.Avg**

4.4.3 Network Monitoring

You can observe the performance counters listed in [4.4.2 Activation Verification](#) to monitor the service status of RFSP groups in a cell after resource control in NSA networking is enabled.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Flexible User Steering*
- *Network Slicing*
- *Multi-Operator Sharing*
- *Cell Combination*
- *FWA*
- *URLLC*
- *UL and DL Decoupling*
- *Super Uplink*
- *QCI-based Resource Control*
- *Cell Management*
- *Service-differentiated Experience Guarantee*
- *Hyper Cell*